Project 02 — Multi-Site Expansion + DHCP

Branch Switching · WAN · Centralized DHCP · IPv6 Dual-Stack · DNS

Series	Enterprise Network Labs — CML 2.9
Project	02 of 13
Platform	Cisco Modeling Labs 2.9 (IOSv + IOL-L2)
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Purpose	CCNA 200-301 hands-on lab documentation

Overview

HQ exists as an isolated campus. A second site (branch) opens. Build the branch from scratch, connect over a point-to-point WAN, deploy centralized DHCP serving both sites via relay agent, add IPv6 dual-stack with SLAAC, and verify DNS end-to-end across the WAN.

STAR Summary

Situation	Single-site campus with no branch, no DHCP, no DNS. All endpoints using static IPs.
Task	Build branch site, WAN link, centralized DHCP with relay, IPv6 dual-stack, DNS resolution.
Action	Deployed BR-RTR1/DSW1/ASW1, /30 WAN, static routes, Dnsmasq with relay pools, IPv6, Dnsmasq DNS.
Result	Multi-site network with centralized DHCP serving 8 VLANs across 2 sites, IPv6 SLAAC, cross-site DNS.

Lab Topology

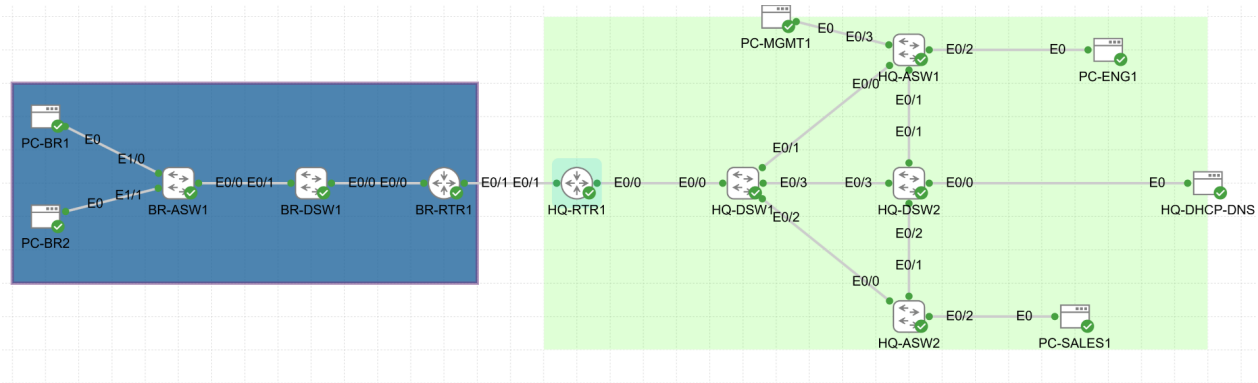

Figure 1 — CML Lab Topology (Cisco Modeling Labs 2.9)

Technologies Used

- Router-on-a-Stick inter-VLAN routing at branch site (BR-RTR1)
- 802.1Q trunking with VLAN pruning — VLANs 100, 200, 300, 500, 999, 1000
- Point-to-point WAN /30 link (10.0.0.0/30) between HQ-RTR1 and BR-RTR1
- Static routing — explicit per-subnet routes in both directions
- Centralized DHCP (Dnsmasq on HQ-DHCP-DNS at 10.1.99.50)
- ip helper-address DHCP relay on all router subinterfaces at both sites
- DHCP pools for 8 VLANs across 2 sites with static MAC reservations
- IPv6 dual-stack — /126 WAN, /64 VLAN 100 at both sites
- SLAAC (Stateless Address Autoconfiguration) via Router Advertisements
- DNS with static A records (Dnsmasq address= directives)
- Rapid-PVST+ with BR-DSW1 as root for all branch VLANs

- PortFast + BPDU Guard on all branch access ports
- SSH v2 with VTY ACL restricted to management VLAN 999
- ip routing on BR-ASW1 for remote-subnet ICMP reply routing
- bandwidth/delay tuned on WAN interfaces for Project 03 OSPF cost accuracy

IP Addressing

Branch Site (10.2.x.x)

Device	Interface	IP Address	Purpose
BR-RTR1	E0/0.100	10.2.100.1/24	Engineering gateway
BR-RTR1	E0/0.200	10.2.200.1/24	Sales gateway
BR-RTR1	E0/0.300	10.2.44.1/24	Guest gateway
BR-RTR1	E0/0.500	10.2.50.1/24	Voice gateway
BR-RTR1	E0/0.999	10.2.99.1/24	Management gateway
BR-DSW1	Vlan999	10.2.99.2/24	Management SVI
BR-ASW1	Vlan999	10.2.99.3/24	Management SVI
PC-BR1	eth0	10.2.100.197/24	DHCP reserved (Engineering)
PC-BR2	eth0	10.2.200.108/24	DHCP reserved (Sales)

WAN Link + HQ Additions

Device	Interface	IP Address	Purpose
HQ-RTR1	E0/1	10.0.0.1/30	WAN hub-side
BR-RTR1	E0/1	10.0.0.2/30	WAN spoke-side
HQ-RTR1	E0/0.500	10.1.50.1/24	HQ Voice gateway (new)
HQ-DHCP-DNS	eth0	10.1.99.50/24	Centralized DHCP + DNS server

IPv6 Addressing

Interface	IPv6 Address	Purpose
HQ-RTR1 E0/1	2001:db8:0:1::1/126	WAN HQ-side
BR-RTR1 E0/1	2001:db8:0:1::2/126	WAN Branch-side
HQ-RTR1 E0/0.100	2001:db8:1:100::1/64	HQ Engineering IPv6 GW
BR-RTR1 E0/0.100	2001:db8:2:100::1/64	Branch Engineering IPv6 GW
PC-BR1 eth0	2001:db8:2:100:5054:ff:fe22:a69c/64	SLAAC (EUI-64 from MAC)

Phase 1 — Branch Site Base

BR-RTR1 — Router-on-a-Stick

WHY: Parent interface carries all VLAN traffic — no IP assigned. encapsulation dot1Q selects which VLAN tag this subinterface processes. VLAN 500 Voice built now to avoid rework in Project 11 QoS phase.

```
interface Ethernet0/0
 description TRUNK-TO-BR-DSW1-E0/0
 no ip address
 no shutdown

interface Ethernet0/0.100
 description GATEWAY-VLAN100-ENGINEERING
 encapsulation dot1Q 100
 ip address 10.2.100.1 255.255.255.0
 no shutdown

interface Ethernet0/0.200
 description GATEWAY-VLAN200-SALES
 encapsulation dot1Q 200
 ip address 10.2.200.1 255.255.255.0
 no shutdown

interface Ethernet0/0.300
 description GATEWAY-VLAN300-GUEST
 encapsulation dot1Q 300
 ip address 10.2.44.1 255.255.255.0
 no shutdown

interface Ethernet0/0.500
 description GATEWAY-VLAN500-VOICE
 encapsulation dot1Q 500
 ip address 10.2.50.1 255.255.255.0
 no shutdown

interface Ethernet0/0.999
 description GATEWAY-VLAN999-MANAGEMENT
 encapsulation dot1Q 999
 ip address 10.2.99.1 255.255.255.0
 no shutdown

interface Ethernet0/0.1000
 description NATIVE-VLAN1000-UNUSED
 encapsulation dot1Q 1000 native
 no shutdown
```

BR-DSW1 — STP Root + Trunking

WHY: ip routing enabled now — Project 03 needs DSW1 as an OSPF area router. encapsulation dot1q must precede switchport mode trunk on IOL-L2. Explicit VLAN allowed lists prevent accidental VLAN leakage.

```
ip routing
spanning-tree mode rapid-pvst
spanning-tree vlan 100,200,300,500,999,1000 priority 4096

interface Ethernet0/0
 description TRUNK-TO-BR-RTR1-E0/0
```

```
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 1000
 switchport trunk allowed vlan 100,200,300,500,999,1000
 no shutdown

interface Ethernet0/1
 description TRUNK-TO-BR-ASW1-E0/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk native vlan 1000
 switchport trunk allowed vlan 100,200,300,500,999,1000
 no shutdown
```

BR-ASW1 — Access Ports + ip routing

WHY: ip routing enabled on BR-ASW1 — ip default-gateway alone cannot route replies to remote-subnet sources (e.g. HQ-RTR1 pinging from 10.0.0.1). Static default route ensures all unknown destinations reach BR-RTR1.

```
ip routing
ip route 0.0.0.0 0.0.0.0 10.2.99.1

interface Ethernet1/0
 description ACCESS-PC-BR1-VLAN100
 switchport mode access
 switchport access vlan 100
 switchport nonegotiate
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown

interface Ethernet1/1
 description ACCESS-PC-BR2-VLAN200
 switchport mode access
 switchport access vlan 200
 switchport nonegotiate
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown
```

Phase 1 Verification Screenshots

```
BR-RTR1#sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
Ethernet0/0            unassigned      YES TFTP   up                    up
Ethernet0/0.100        10.2.100.1      YES manual up                    up
Ethernet0/0.200        10.2.200.1      YES manual up                    up
Ethernet0/0.300        10.2.44.1       YES manual up                    up
Ethernet0/0.500        10.2.50.1       YES manual up                    up
Ethernet0/0.999        10.2.99.1       YES manual up                    up
Ethernet0/0.1000       unassigned      YES unset  up                    up
Ethernet0/1            unassigned      YES TFTP   administratively down down
Ethernet0/2            unassigned      YES TFTP   administratively down down
Ethernet0/3            unassigned      YES TFTP   administratively down down
BR-RTR1#
```

BR-RTR1 show ip interface brief — all subinterfaces E0/0.100–.999 up/up

```
BR-RTR1#sh cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID        Local Intrfce     Holdtme    Capability  Platform  Port ID
BR-DSW1.lab.local
                 Eth 0/0           161             R S I  Linux Uni Eth 0/0

Total cdp entries displayed : 1
BR-RTR1#
```

BR-RTR1 show cdp neighbors — BR-DSW1 confirmed on E0/0

```
BR-DSW1#show interfaces trunk

Port          Mode             Encapsulation  Status        Native vlan
Et0/0         on               802.1q         trunking      1000
Et0/1         on               802.1q         trunking      1000

Port          Vlans allowed on trunk
Et0/0         100,200,300,500,999-1000
Et0/1         100,200,300,500,999-1000

Port          Vlans allowed and active in management domain
Et0/0         100,200,300,500,999-1000
Et0/1         100,200,300,500,999-1000

Port          Vlans in spanning tree forwarding state and not pruned
Et0/0         100,200,300,500,999-1000
Et0/1         100,200,300,500,999-1000
```

BR-DSW1 show interfaces trunk — E0/0 and E0/1 trunking, native 1000

```
BR-DSW1#show spanning-tree vlan 100

VLAN0100
  Spanning tree enabled protocol rstp
  Root ID    Priority    4196
             Address     aabb.cc00.0f00
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    4196   (priority 4096 sys-id-ext 100)
             Address     aabb.cc00.0f00
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- -------------------------------
Et0/0               Desg FWD 100       128.1    P2p
Et0/1               Desg FWD 100       128.2    P2p
```

BR-DSW1 show spanning-tree vlan 100 — 'This bridge is the root', priority 4096

```
BR-ASW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Et0/1, Et0/2, Et0/3, Et1/2
                                                Et1/3
100  ENGINEERING                      active    Et1/0
200  SALES                            active    Et1/1
300  GUEST                            active
500  VOICE                            active
999  MANAGEMENT                       active
1000 NATIVE-UNUSED                    active
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
```

BR-ASW1 show vlan brief — PC-BR1 in VLAN 100, PC-BR2 in VLAN 200

```
BR-ASW1# show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID        Local Intrfce     Holdtme    Capability  Platform  Port ID
BR-DSW1.lab.local
                 Eth 0/0           172             R S I  Linux Uni Eth 0/1

Total cdp entries displayed : 1
BR-ASW1#ping 10.2.99.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.2.99.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/4 ms
BR-ASW1#
```

Cross-device ping — management plane reachable across all branch devices

Phase 2 — WAN Connectivity

HQ-RTR1 — WAN Interface + Static Routes to Branch

WHY: /30 gives exactly 2 usable IPs — correct for point-to-point links. bandwidth 1000 and delay 1000 set now for OSPF cost accuracy in Project 03 (IOL defaults to 10 Mbps which will produce wrong OSPF costs). Explicit per-subnet routes give better visibility than a summary during troubleshooting.

```
interface Ethernet0/1
 description WAN-TO-BR-RTR1-E0/1
 ip address 10.0.0.1 255.255.255.252
 bandwidth 1000
 delay 1000
 no shutdown

ip route 10.2.100.0 255.255.255.0 10.0.0.2
ip route 10.2.200.0 255.255.255.0 10.0.0.2
ip route 10.2.44.0  255.255.255.0 10.0.0.2
ip route 10.2.50.0  255.255.255.0 10.0.0.2
ip route 10.2.99.0  255.255.255.0 10.0.0.2
```

BR-RTR1 — WAN Interface + Static Routes to HQ

```
interface Ethernet0/1
 description WAN-TO-HQ-RTR1-E0/1
 ip address 10.0.0.2 255.255.255.252
 bandwidth 1000
 delay 1000
 no shutdown

ip route 10.1.100.0 255.255.255.0 10.0.0.1
ip route 10.1.200.0 255.255.255.0 10.0.0.1
ip route 10.1.44.0  255.255.255.0 10.0.0.1
ip route 10.1.50.0  255.255.255.0 10.0.0.1
ip route 10.1.99.0  255.255.255.0 10.0.0.1
ip route 10.0.255.1 255.255.255.255 10.0.0.1
```

Phase 2 Verification Screenshots

```
Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 16 subnets, 3 masks
C        10.0.0.0/30 is directly connected, Ethernet0/1
L        10.0.0.1/32 is directly connected, Ethernet0/1
C        10.0.255.1/32 is directly connected, Loopback0
C        10.1.44.0/24 is directly connected, Ethernet0/0.300
L        10.1.44.1/32 is directly connected, Ethernet0/0.300
C        10.1.99.0/24 is directly connected, Ethernet0/0.999
L        10.1.99.1/32 is directly connected, Ethernet0/0.999
C        10.1.100.0/24 is directly connected, Ethernet0/0.100
L        10.1.100.1/32 is directly connected, Ethernet0/0.100
C        10.1.200.0/24 is directly connected, Ethernet0/0.200
L        10.1.200.1/32 is directly connected, Ethernet0/0.200
S        10.2.44.0/24 [1/0] via 10.0.0.2
S        10.2.50.0/24 [1/0] via 10.0.0.2
S        10.2.99.0/24 [1/0] via 10.0.0.2
S        10.2.100.0/24 [1/0] via 10.0.0.2
S        10.2.200.0/24 [1/0] via 10.0.0.2
```

HQ-RTR1 show ip route — 5 static routes to 10.2.x.x branch subnets

```
Connection to 10.1.99.11 closed.
PC-MGMT1:~# ping 10.2.99.3
PING 10.2.99.3 (10.2.99.3): 56 data bytes
64 bytes from 10.2.99.3: seq=737 ttl=253 time=7.757 ms
64 bytes from 10.2.99.3: seq=738 ttl=253 time=7.729 ms
64 bytes from 10.2.99.3: seq=739 ttl=253 time=7.397 ms
64 bytes from 10.2.99.3: seq=740 ttl=253 time=7.463 ms
64 bytes from 10.2.99.3: seq=741 ttl=253 time=6.963 ms
64 bytes from 10.2.99.3: seq=742 ttl=253 time=6.780 ms
64 bytes from 10.2.99.3: seq=743 ttl=253 time=7.065 ms
64 bytes from 10.2.99.3: seq=744 ttl=253 time=7.316 ms
64 bytes from 10.2.99.3: seq=745 ttl=253 time=7.615 ms
64 bytes from 10.2.99.3: seq=746 ttl=253 time=7.132 ms
64 bytes from 10.2.99.3: seq=747 ttl=253 time=7.209 ms
64 bytes from 10.2.99.3: seq=748 ttl=253 time=7.297 ms
64 bytes from 10.2.99.3: seq=749 ttl=253 time=7.724 ms
64 bytes from 10.2.99.3: seq=750 ttl=253 time=7.460 ms
64 bytes from 10.2.99.3: seq=751 ttl=253 time=6.897 ms
64 bytes from 10.2.99.3: seq=752 ttl=253 time=7.473 ms
64 bytes from 10.2.99.3: seq=753 ttl=253 time=8.514 ms
64 bytes from 10.2.99.3: seq=754 ttl=253 time=7.786 ms
64 bytes from 10.2.99.3: seq=755 ttl=253 time=6.703 ms
64 bytes from 10.2.99.3: seq=756 ttl=253 time=7.093 ms
64 bytes from 10.2.99.3: seq=757 ttl=253 time=6.934 ms
64 bytes from 10.2.99.3: seq=758 ttl=253 time=7.664 ms
```

Cross-site ping — PC-MGMT1 (10.1.99.100) reaching BR-ASW1 (10.2.99.3): SUCCESS

```
BR-RTR1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 18 subnets, 3 masks
C        10.0.0.0/30 is directly connected, Ethernet0/1
L        10.0.0.2/32 is directly connected, Ethernet0/1
S        10.1.40.0/24 [1/0] via 10.0.0.1
S        10.1.44.0/24 [1/0] via 10.0.0.1
S        10.1.50.0/24 [1/0] via 10.0.0.1
S        10.1.99.0/24 [1/0] via 10.0.0.1
S        10.1.100.0/24 [1/0] via 10.0.0.1
S        10.1.200.0/24 [1/0] via 10.0.0.1
C        10.2.44.0/24 is directly connected, Ethernet0/0.300
L        10.2.44.1/32 is directly connected, Ethernet0/0.300
C        10.2.50.0/24 is directly connected, Ethernet0/0.500
L        10.2.50.1/32 is directly connected, Ethernet0/0.500
C        10.2.99.0/24 is directly connected, Ethernet0/0.999
L        10.2.99.1/32 is directly connected, Ethernet0/0.999
C        10.2.100.0/24 is directly connected, Ethernet0/0.100
L        10.2.100.1/32 is directly connected, Ethernet0/0.100
C        10.2.200.0/24 is directly connected, Ethernet0/0.200
L        10.2.200.1/32 is directly connected, Ethernet0/0.200
```

BR-RTR1 show ip route — 6 static routes to 10.1.x.x HQ subnets

Phase 3 — Centralized DHCP

HQ-RTR1 + BR-RTR1 — DHCP Relay (ip helper-address)

WHY: helper-address goes on the SUBINTERFACE — not the physical parent interface. The relay agent uses the subinterface IP as the giaddr field in the DHCP packet. Dnsmasq reads giaddr to match the correct DHCP pool. ALL data VLAN subinterfaces need helper-address — missing one = that VLAN never gets DHCP.

```
! On HQ-RTR1 — add helper-address to all data subinterfaces
interface Ethernet0/0.100
 ip helper-address 10.1.99.50
interface Ethernet0/0.200
 ip helper-address 10.1.99.50
interface Ethernet0/0.300
 ip helper-address 10.1.99.50
interface Ethernet0/0.500
 ip helper-address 10.1.99.50

! On BR-RTR1 — same pattern for branch subinterfaces
interface Ethernet0/0.100
 ip helper-address 10.1.99.50
interface Ethernet0/0.200
 ip helper-address 10.1.99.50
interface Ethernet0/0.300
 ip helper-address 10.1.99.50
interface Ethernet0/0.500
 ip helper-address 10.1.99.50
```

HQ-DHCP-DNS — Dnsmasq Pool Configuration

WHY: Dnsmasq reads the giaddr field to select the matching DHCP pool. dhcp-range with set: assigns the pool to the matching subnet. Static reservations lock endpoints to permanent IPs via MAC — required so that address= DNS records remain accurate across reboots.

```
domain=lab.local
local=/lab.local/
log-dhcp

# HQ pools — giaddr matches HQ subinterface IPs
dhcp-range=set:hq-eng,10.1.100.100,10.1.100.200,255.255.255.0,24h
dhcp-option=tag:hq-eng,option:router,10.1.100.1
dhcp-option=tag:hq-eng,option:dns-server,10.1.99.50

dhcp-range=set:hq-sales,10.1.200.100,10.1.200.200,255.255.255.0,24h
dhcp-option=tag:hq-sales,option:router,10.1.200.1
dhcp-option=tag:hq-sales,option:dns-server,10.1.99.50

# Branch pools — giaddr matches BR-RTR1 subinterface IPs
dhcp-range=set:br-eng,10.2.100.100,10.2.100.200,255.255.255.0,24h
dhcp-option=tag:br-eng,option:router,10.2.100.1
dhcp-option=tag:br-eng,option:dns-server,10.1.99.50

dhcp-range=set:br-sales,10.2.200.100,10.2.200.200,255.255.255.0,24h
dhcp-option=tag:br-sales,option:router,10.2.200.1
dhcp-option=tag:br-sales,option:dns-server,10.1.99.50

# Static DHCP reservations by MAC
```

```
dhcp-host=52:54:00:22:a6:9c,pc-br1,10.2.100.197
dhcp-host=52:54:00:53:da:69,pc-br2,10.2.200.108

# DNS A records
address=/hq-rtr1.lab.local/10.1.99.1
address=/br-rtr1.lab.local/10.2.99.1
address=/hq-dhcp-dns.lab.local/10.1.99.50
address=/pc-br1.lab.local/10.2.100.197
address=/pc-br2.lab.local/10.2.200.108
```

Phase 3 Verification Screenshots

```
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 requested options: 1:netmask, 3:router, 6:dns-server, 12:hostname,
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 requested options: 15:domain-name, 28:broadcast, 42:ntp-server
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 next server: 10.1.99.50
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  1 option: 53 message-type  2
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option: 54 server-identifier  10.1.99.50
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option: 51 lease-time  1d
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option: 58 T1  12h
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option: 59 T2  21h
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option:  1 netmask  255.255.255.0
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option: 28 broadcast  10.2.100.255
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option:  6 dns-server  10.1.99.50
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option:  3 router  10.2.100.1
Apr  5 20:15:44 dnsmasq-dhcp[1]: no address range available for DHCP request via eth0
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 available DHCP range: 10.2.100.100 -- 10.2.100.200
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 vendor class: udhcp 1.37.0
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 client provides name: localhost
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 DHCPREQUEST(eth0) 10.2.100.197 52:54:00:22:a6:9c
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 tags: br-eng, eth0
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 DHCPACK(eth0) 10.2.100.197 52:54:00:22:a6:9c localhost
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 requested options: 1:netmask, 3:router, 6:dns-server, 12:hostname,
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 requested options: 15:domain-name, 28:broadcast, 42:ntp-server
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 next server: 10.1.99.50
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  1 option: 53 message-type  5
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option: 54 server-identifier  10.1.99.50
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option: 51 lease-time  1d
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option: 58 T1  12h
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option: 59 T2  21h
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option:  1 netmask  255.255.255.0
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option: 28 broadcast  10.2.100.255
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  9 option: 12 hostname  localhost
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option:  6 dns-server  10.1.99.50
Apr  5 20:15:44 dnsmasq-dhcp[1]: 2800685166 sent size:  4 option:  3 router  10.2.100.1
Apr  5 20:15:44 dnsmasq-dhcp[1]: not giving name localhost to the DHCP lease of 10.2.100.197 because the name exis
Apr  5 20:15:45 dnsmasq-dhcp[1]: no address range available for DHCP request via eth0
```

dnsmasq log — live DHCP lease assignments from both HQ and Branch

```
BR-RTR1#show ip arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  10.0.0.1               174   aabb.cc00.0910  ARPA   Ethernet0/1
Internet  10.0.0.2                 -   aabb.cc00.1010  ARPA   Ethernet0/1
Internet  10.2.44.1                -   aabb.cc00.1000  ARPA   Ethernet0/0.300
Internet  10.2.50.1                -   aabb.cc00.1000  ARPA   Ethernet0/0.500
Internet  10.2.99.1                -   aabb.cc00.1000  ARPA   Ethernet0/0.999
Internet  10.2.99.2               46   aabb.cc80.0f00  ARPA   Ethernet0/0.999
Internet  10.2.99.3               13   aabb.cc80.0e00  ARPA   Ethernet0/0.999
Internet  10.2.100.1               -   aabb.cc00.1000  ARPA   Ethernet0/0.100
Internet  10.2.100.197             6   5254.0022.a69c  ARPA   Ethernet0/0.100
Internet  10.2.200.1               -   aabb.cc00.1000  ARPA   Ethernet0/0.200
Internet  10.2.200.108            44   5254.0053.da69  ARPA   Ethernet0/0.200
BR-RTR1#
```

BR-RTR1 show ip arp — PC-BR1 (10.2.100.197) and PC-BR2 (10.2.200.108) leased

```
HQ-RTR1#
HQ-RTR1#show ip arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  10.0.0.1                -   aabb.cc00.0910  ARPA   Ethernet0/1
Internet  10.0.0.2              175   aabb.cc00.1010  ARPA   Ethernet0/1
Internet  10.1.44.1               -   aabb.cc00.0900  ARPA   Ethernet0/0.300
Internet  10.1.99.1               -   aabb.cc00.0900  ARPA   Ethernet0/0.999
Internet  10.1.99.11             79   aabb.cc80.0a00  ARPA   Ethernet0/0.999
Internet  10.1.99.14              7   aabb.cc80.0d00  ARPA   Ethernet0/0.999
Internet  10.1.99.50              7   5254.00f2.2230  ARPA   Ethernet0/0.999
Internet  10.1.99.100             0   5254.00b2.8d53  ARPA   Ethernet0/0.999
Internet  10.1.100.1              -   aabb.cc00.0900  ARPA   Ethernet0/0.100
Internet  10.1.100.10             0   5254.00d7.cbbc  ARPA   Ethernet0/0.100
Internet  10.1.100.194           46   5254.00d7.cbbc  ARPA   Ethernet0/0.100
Internet  10.1.200.1              -   aabb.cc00.0900  ARPA   Ethernet0/0.200
Internet  10.1.200.10            22   5254.001e.faaf  ARPA   Ethernet0/0.200
Internet  10.1.200.142           46   5254.001e.faaf  ARPA   Ethernet0/0.200
HQ-RTR1#
```

HQ-RTR1 show ip arp — HQ endpoints with DHCP leases visible

```
-sh: show: not found
inserthostname-here:~# ip addr show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP qlen 1000
    link/ether 52:54:00:53:da:69 brd ff:ff:ff:ff:ff:ff
    inet 10.2.200.108/24 brd 10.2.200.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::5054:ff:fe53:da69/64 scope link
       valid_lft forever preferred_lft forever
inserthostname-here:~#
```

PC-BR2 ip addr show eth0 — 10.2.200.108 leased from 10.1.99.50

```
HQ-DSW2#sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
100  Engineering                      active
200  Sales                            active
300  Guest                            active
999  Management                       active    Et0/0
1000 NATIVE-UNUSED                    active
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
HQ-DSW2#
```

HQ-DSW2 show vlan brief — DHCP server port Et0/0 confirmed in VLAN 999

Phase 4 — IPv6 Dual-Stack

HQ-RTR1 + BR-RTR1 — IPv6 Configuration

WHY: ipv6 unicast-routing must be enabled globally — without it the router drops all IPv6 packets. /64 prefix is mandatory for SLAAC (hosts build their full address from the /64 prefix + EUI-64). PC-BR1 EUI-64 from MAC 52:54:00:22:a6:9c → insert ff:fe → flip 7th bit → 5054:ff:fe22:a69c.

```
! HQ-RTR1
ipv6 unicast-routing

interface Ethernet0/1
 ipv6 address 2001:db8:0:1::1/126
 ipv6 enable

interface Ethernet0/0.100
 ipv6 address 2001:db8:1:100::1/64
 ipv6 enable

ipv6 route 2001:db8:2:100::/64 2001:db8:0:1::2

! BR-RTR1
ipv6 unicast-routing

interface Ethernet0/1
 ipv6 address 2001:db8:0:1::2/126
 ipv6 enable

interface Ethernet0/0.100
 ipv6 address 2001:db8:2:100::1/64
 ipv6 enable

ipv6 route 2001:db8:1:100::/64 2001:db8:0:1::1
```

Phase 4 Verification Screenshots

```
HQ-RTR1#show ipv6 interface brief
Ethernet0/0            [up/up]
    unassigned
Ethernet0/0.100        [up/up]
    FE80::A8BB:CCFF:FE00:900
    2001:DB8:1:100::1
Ethernet0/0.200        [up/up]
    unassigned
Ethernet0/0.300        [up/up]
    unassigned
Ethernet0/0.500        [up/up]
    unassigned
Ethernet0/0.999        [up/up]
    unassigned
Ethernet0/1            [up/up]
    FE80::A8BB:CCFF:FE00:910
    2001:DB8:0:1::1
Ethernet0/2            [administratively down/down]
    unassigned
Ethernet0/3            [administratively down/down]
    unassigned
Loopback0              [up/up]
    unassigned
```

HQ-RTR1 show ipv6 route — static route S 2001:db8:2:100::/64 via 2001:db8:0:1::2

```
BR-RTR1#show ipv6 interface brief
Ethernet0/0            [up/up]
    unassigned
Ethernet0/0.100        [up/up]
    FE80::A8BB:CCFF:FE00:1000
    2001:DB8:2:100::1
Ethernet0/0.200        [up/up]
    unassigned
Ethernet0/0.300        [up/up]
    unassigned
Ethernet0/0.500        [up/up]
    unassigned
Ethernet0/0.999        [up/up]
    unassigned
Ethernet0/0.1000       [up/up]
    unassigned
Ethernet0/1            [up/up]
    FE80::A8BB:CCFF:FE00:1010
    2001:DB8:0:1::2
Ethernet0/2            [administratively down/down]
    unassigned
Ethernet0/3            [administratively down/down]
    unassigned
BR-RTR1#show ipv6 route
```

BR-RTR1 show ipv6 route — static route S 2001:db8:1:100::/64 via 2001:db8:0:1::1

```
HQ-RTR1#ping ipv6 2001:db8:0:1::2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:0:1::2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
HQ-RTR1#ping ipv6 2001:db8:2:100::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:2:100::1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
HQ-RTR1#ping 2001:db8:2:100::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:2:100::1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
HQ-RTR1#
```

Cross-site IPv6 ping — HQ-RTR1 to BR-RTR1 VLAN 100 (2001:db8:2:100::1): SUCCESS

```
inserthostname-here:~# ip -6 addr show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UP qlen 1000
    inet6 2001:db8:2:100:5054:ff:fe22:a69c/64 scope global dynamic flags 100
       valid_lft 2591864sec preferred_lft 604664sec
    inet6 fe80::5054:ff:fe22:a69c/64 scope link
       valid_lft forever preferred_lft forever
inserthostname-here:~# ping6 2001:db8:2:100::1
PING 2001:db8:2:100::1 (2001:db8:2:100::1): 56 data bytes
64 bytes from 2001:db8:2:100::1: seq=0 ttl=64 time=8.380 ms
64 bytes from 2001:db8:2:100::1: seq=1 ttl=64 time=3.681 ms
64 bytes from 2001:db8:2:100::1: seq=2 ttl=64 time=4.135 ms
64 bytes from 2001:db8:2:100::1: seq=3 ttl=64 time=3.834 ms
64 bytes from 2001:db8:2:100::1: seq=4 ttl=64 time=4.427 ms
64 bytes from 2001:db8:2:100::1: seq=5 ttl=64 time=3.992 ms
64 bytes from 2001:db8:2:100::1: seq=6 ttl=64 time=3.475 ms
64 bytes from 2001:db8:2:100::1: seq=7 ttl=64 time=3.822 ms
64 bytes from 2001:db8:2:100::1: seq=8 ttl=64 time=4.103 ms
64 bytes from 2001:db8:2:100::1: seq=9 ttl=64 time=4.275 ms
64 bytes from 2001:db8:2:100::1: seq=10 ttl=64 time=3.653 ms
64 bytes from 2001:db8:2:100::1: seq=11 ttl=64 time=3.931 ms
64 bytes from 2001:db8:2:100::1: seq=12 ttl=64 time=3.770 ms
64 bytes from 2001:db8:2:100::1: seq=13 ttl=64 time=4.467 ms
64 bytes from 2001:db8:2:100::1: seq=14 ttl=64 time=4.304 ms
64 bytes from 2001:db8:2:100::1: seq=15 ttl=64 time=3.481 ms
^C
--- 2001:db8:2:100::1 ping statistics ---
16 packets transmitted, 16 packets received, 0% packet loss
round-trip min/avg/max = 3.475/4.233/8.380 ms
inserthostname-here:~# ip -6 addr show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UP qlen 1000
    inet6 2001:db8:2:100:5054:ff:fe22:a69c/64 scope global dynamic flags 100
       valid_lft 2591915sec preferred_lft 604715sec
    inet6 fe80::5054:ff:fe22:a69c/64 scope link
       valid_lft forever preferred_lft forever
inserthostname-here:~#
```

PC-BR1 ip -6 addr show eth0 — SLAAC address 2001:db8:2:100:5054:ff:fe22:a69c/64

Phase 5 — DNS End-to-End

Router DNS Client Configuration

WHY: no ip domain lookup was set during hardening to prevent IOS from DNS-resolving mistyped commands (causes a 30-second timeout). Re-enable it now that we have a valid name-server. ip domain lookup + ip name-server lets routers resolve FQDNs.

```
! On HQ-RTR1 and BR-RTR1
ip domain lookup
ip name-server 10.1.99.50
```

Phase 5 Verification Screenshots

```
** server can't find hq-rtr1.lab.local: NXDOMAIN

pc-br1:~# nslookup br-rtr1.lab.local 10.1.99.50
Server:         10.1.99.50
Address:        10.1.99.50:53

Name:   br-rtr1.lab.local
Address: 10.2.99.1

** server can't find br-rtr1.lab.local: NXDOMAIN

pc-br1:~# nslookup hq-dhcp-dns.lab.local 10.1.99.50
Server:         10.1.99.50
Address:        10.1.99.50:53

Name:   hq-dhcp-dns.lab.local
Address: 10.1.99.50

** server can't find hq-dhcp-dns.lab.local: NXDOMAIN

pc-br1:~# nslookup pc-br1.lab.local 10.1.99.50
Server:         10.1.99.50
Address:        10.1.99.50:53

** server can't find pc-br1.lab.local: NXDOMAIN

** server can't find pc-br1.lab.local: NXDOMAIN

pc-br1:~# ping -c 3 10.1.99.1
PING 10.1.99.1 (10.1.99.1): 56 data bytes
64 bytes from 10.1.99.1: seq=0 ttl=254 time=4.966 ms
64 bytes from 10.1.99.1: seq=1 ttl=254 time=4.693 ms
64 bytes from 10.1.99.1: seq=2 ttl=254 time=4.672 ms

--- 10.1.99.1 ping statistics ---
```

PC-BR1 nslookup hq-rtr1.lab.local — resolves to 10.1.99.1 via 10.1.99.50

```
pc-br2:~#
pc-br2:~#
pc-br2:~# nslookup pc-br1.lab.local 10.1.99.50
Server:         10.1.99.50
Address:        10.1.99.50:53

** server can't find pc-br1.lab.local: NXDOMAIN

** server can't find pc-br1.lab.local: NXDOMAIN

pc-br2:~# nslookup pc-br1.lab.local 10.1.99.50
Server:         10.1.99.50
Address:        10.1.99.50:53

** server can't find pc-br1.lab.local: NXDOMAIN

** server can't find pc-br1.lab.local: NXDOMAIN

pc-br2:~# nslookup hq-rtr1.lab.local 10.1.99.50
Server:         10.1.99.50
Address:        10.1.99.50:53

Name:   hq-rtr1.lab.local
Address: 10.1.99.1

** server can't find hq-rtr1.lab.local: NXDOMAIN

pc-br2:~#
```

PC-BR1 nslookup br-rtr1.lab.local — resolves to 10.2.99.1

```
BR-RTR1#show running-config | include name-server
ip name-server 10.1.99.50
BR-RTR1#show running-config | include domain
ip domain name lab.local
BR-RTR1#ping hq-rtr1.lab.local
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.99.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
BR-RTR1#ping hq-dhcp-dns.lab.local
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.99.50, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/3/4 ms
BR-RTR1#
```

BR-RTR1 ping hq-rtr1.lab.local — name resolves cross-site, ping succeeds

```
HQ-RTR1#show running-config | include name-server
ip name-server 10.1.99.50
HQ-RTR1#show running-config | include domain
ip domain name lab.local
HQ-RTR1#ping br-rtr1.lab.local
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.2.99.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
HQ-RTR1#ping pc-br1.lab.local
```

HQ-RTR1 ping br-rtr1.lab.local — name resolves, cross-site ping succeeds

```
HQ-DSW2#show interfaces Ethernet0/0 switchport
Name: Et0/0
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: negotiate
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 999 (Management)
Trunking Native Mode VLAN: 1 (default)
Administrative Native VLAN tagging: disabled
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk Native VLAN tagging: enabled
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk associations: none
Administrative private-vlan trunk mappings: none
Operational private-vlan: none
Trunking VLANs Enabled: ALL
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL

Protected: false
Appliance trust: none
HQ-DSW2#
```

HQ-DSW2 show vlan brief — Et0/0 confirmed in VLAN 999 for DHCP-DNS server

Break/Fix Challenge — Three Simultaneous DHCP Faults

Three simultaneous DHCP faults are introduced: (1) wrong helper-address on BR-RTR1 subinterfaces pointing to a non-existent IP, (2) helper-address removed from HQ-RTR1 VLAN 100 subinterface, (3) DHCP server process stopped on HQ-DHCP-DNS. All three must be found and fixed simultaneously.

Diagnosis Approach

```
! 1. Check if PC-BR1 gets DHCP — udhcpc -i eth0 (times out = relay or server fault)
! 2. show run | section helper on BR-RTR1 — verify helper-address is 10.1.99.50
! 3. show run | section helper on HQ-RTR1 — verify all subinterfaces have helper
! 4. SSH to HQ-DHCP-DNS — check: systemctl status dnsmasq or ps aux | grep dnsmasq
! 5. Fix all three, then re-run udhcpc on PC-BR1 and PC-BR2
```

Break/Fix Screenshots

```
BR-RTR1#show ip route 10.1.99.0
Routing entry for 10.1.99.0/24
  Known via "static", distance 1, metric 0
  Routing Descriptor Blocks:
  * 10.0.0.1
      Route metric is 0, traffic share count is 1
BR-RTR1#show running-config | section helper
 ip helper-address 10.1.99.99
 ip helper-address 10.1.99.50
 ip helper-address 10.1.99.50
 ip helper-address 10.1.99.50
BR-RTR1#enable
BR-RTR1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
BR-RTR1(config)#interface Ethernet0/0.100
BR-RTR1(config-subif)# no ip helper-address 10.1.99.99
BR-RTR1(config-subif)# ip helper-address 10.1.99.50
BR-RTR1(config-subif)#end
BR-RTR1#write memory
*Apr  6 00:00:21.748: %SYS-5-CONFIG_I: Configured from console by console
BR-RTR1#write memory
Building configuration...
[OK]
BR-RTR1#show running-config | section helper
 ip helper-address 10.1.99.50
 ip helper-address 10.1.99.50
 ip helper-address 10.1.99.50
 ip helper-address 10.1.99.50
BR-RTR1#
```

BR-RTR1 fixed — all 4 subinterfaces show ip helper-address 10.1.99.50

```
HQ-RTR1#no debug ip dhcp server packet
DHCP server packet debugging is off.
HQ-RTR1#ping 10.1.99.50
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.99.50, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/3 ms
HQ-RTR1#show running-config | section helper
 ip helper-address 10.1.99.50
 ip helper-address 10.1.99.99
 ip helper-address 10.1.99.50
 ip helper-address 10.1.99.50
HQ-RTR1#enable
HQ-RTR1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
HQ-RTR1(config)#interface Ethernet0/0.200
HQ-RTR1(config-subif)# no ip helper-address 10.1.99.99
HQ-RTR1(config-subif)# ip helper-address 10.1.99.50
HQ-RTR1(config-subif)#end
HQ-RTR1#write memory
Building configuration...
[OK]
HQ-RTR1#
*Apr  6 00:00:33.964: %SYS-5-CONFIG_I: Configured from console by admin on console
HQ-RTR1#show running-config | section helper
 ip helper-address 10.1.99.50
 ip helper-address 10.1.99.50
 ip helper-address 10.1.99.50
 ip helper-address 10.1.99.50
HQ-RTR1#
```

HQ-RTR1 fixed — helper-address restored on VLAN 100 subinterface

```
udhcpc: broadcasting discover
udhcpc: broadcasting discover
udhcpc: broadcasting discover
udhcpc: broadcasting discover
udhcpc: broadcasting discover
udhcpc failed to get a DHCP lease
^C
pc-br1:~# udhcpc -i eth0
udhcpc: started, v1.37.0
udhcpc: broadcasting discover
udhcpc: broadcasting select for 10.2.100.197, server 10.1.99.50
udhcpc: lease of 10.2.100.197 obtained from 10.1.99.50, lease time 86400
pc-br1:~#
```

PC-BR1 DHCP restored — 10.2.100.197 leased from 10.1.99.50 after all three fixes

```
pc-br2:~# udhcpc -i eth0
udhcpc: started, v1.37.0
udhcpc: broadcasting discover
udhcpc: broadcasting select for 10.2.200.108, server 10.1.99.50
udhcpc: lease of 10.2.200.108 obtained from 10.1.99.50, lease time 86400
pc-br2:~#  nslookup pc-br1.lab.local 10.1.99.50
Server:         10.1.99.50
Address:        10.1.99.50:53
```

PC-BR2 DHCP restored — 10.2.200.108 leased from 10.1.99.50

Key Verification Checklist

Command	Device	Expected Result
show ip interface brief	BR-RTR1	E0/0.100–.999 all up/up with 10.2.x.x IPs
show interfaces trunk	BR-DSW1	E0/0 and E0/1 trunking, native 1000, VLANs 100/200/300/500/999
show vlan brief	BR-ASW1	Et1/0 in VLAN 100, Et1/1 in VLAN 200
show spanning-tree vlan 100	BR-DSW1	'This bridge is the root' priority 4196
show ip route	HQ-RTR1	5 static routes to 10.2.x.x via 10.0.0.2
show ip route	BR-RTR1	6 static routes to 10.1.x.x via 10.0.0.1
show run | section helper	Both routers	ip helper-address 10.1.99.50 on all data subinterfaces
udhcpc -i eth0	PC-BR1	Lease 10.2.100.197 from 10.1.99.50
show ipv6 route	HQ-RTR1	S 2001:db8:2:100::/64 via 2001:db8:0:1::2
ip -6 addr show eth0	PC-BR1	SLAAC 2001:db8:2:100:5054:ff:fe22:a69c/64
nslookup hq-rtr1.lab.local 10.1.99.50	PC-BR1	A record: 10.1.99.1
ping br-rtr1.lab.local	HQ-RTR1	Name resolves, cross-site ping succeeds